

# CMVFT: A Multiscale Attention-Guided Framework for Enhanced Keratoconus Suspect Classification in Multiview Corneal Topography



YIFAN LU, BAOJIANG LI, YUNHAI ZHANG, YAYU QI, AND XINGBIN SHI

- **OBJECTIVE:** To develop a multiview fusion framework that effectively identifies suspect keratoconus cases and facilitates the possibility of early clinical intervention.
- **DESIGN:** Retrospective cross-sectional study.
- **SUBJECTS:** A total of 573 corneal topography maps representing eyes classified as normal, suspect, or keratoconus.
- **METHODS:** We designed the corneal multiview fusion transformer (CMVFT), which integrates features from seven standard corneal topography maps. A pretrained ResNet-50 extracts single-view representations that are further refined by a custom-designed multiscale attention module (MSAM). This integrated design specifically compensates for the representation gap commonly encountered when applying Transformers to small-sample corneal topography datasets by dynamically bridging local convolution-based feature extraction with global self-attention mechanisms. A subsequent fusion Transformer then models long-range dependencies across views for comprehensive multiview feature integration.
- **MAIN OUTCOME MEASURES:** The primary measure was the framework's ability to differentiate suspect cases from normal and keratoconus cases, thereby creating a pathway for early clinical intervention.
- **RESULTS:** Experimental evaluation demonstrated that CMVFT effectively distinguishes suspect cases within a feature space characterized by overlapping attributes. Ablation studies confirmed that both the MSAM and the fusion Transformer are essential for robust multiview feature integration, successfully compensating for potential representation shortcomings in small datasets.
- **CONCLUSIONS:** This study is the first to apply a Transformer-driven multiview fusion approach in corneal topography analysis. By compensating for the represen-

tation gap inherent in small-sample settings, CMVFT shows promise in enabling the identification of suspect keratoconus cases and supporting early intervention strategies, with prospective implications for early clinical intervention. (Am J Ophthalmol 2025;280: 87–105. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

## INTRODUCTION

**K**ERATOCONUS (KCN) IS A PROGRESSIVE CORNEAL disorder characterized by corneal protrusion, localized thinning, and irregular astigmatism. If left undiagnosed or untreated, it can progress to severe vision loss.<sup>1–3</sup> In recent years, the global incidence of KCN has increased significantly, highlighting the imperative of early diagnosis and intervention to prevent the disease from advancing toward invasive treatments such as corneal transplantation.<sup>4</sup> Given the growing demand for refractive surgery, there is now an urgent clinical need for high-precision diagnostic methods that address multiclass conditions. One particularly crucial task is to accurately identify suspect KCN cases before surgery, which can prevent iatrogenic corneal ectasia and unwarranted surgical complications.<sup>5</sup> Consequently, the development of robust technologies for suspect KCN detection has substantial clinical relevance: it not only facilitates early intervention but also helps maintain refractive stability within acceptable ranges through simpler correction methods (eg, corrective lenses).

Traditional binary classification approaches (normal vs KCN) often achieve decent accuracy—mainly because normal and advanced-KCN phenotypes exhibit large morphological differences. However, their performance drops sharply in three-class scenarios (normal, suspect, and KCN), where the suspect category's subtle features overlap with either normal or fully manifested KCN, resulting in low sensitivity.<sup>6–9</sup> Moreover, while multiview fusion strategies (eg, simple concatenation, decision-level voting) can leverage multiple corneal topography maps, they fre-

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From the School of Electrical Engineering (Y.L., B.L., X.S.), Shanghai Dianji University, Shanghai, China; Chinese Academy of Sciences (Y.Z), Suzhou Institute of Biomedical Engineering and Technology, Suzhou, China; Department of Endocrinology, Jincheng People's Hospital (Y.Q), Jincheng, China

Inquiries to Baojiang Li, School of Electrical Engineering, Shanghai Dianji University, Shanghai, China.; e-mail: libj@sdju.edu.cn

Inquiries to Yunhai Zhang, Chinese Academy of Sciences, Suzhou Institute of Biomedical Engineering and Technology, Suzhou, China.; e-mail: zhangyh@sibet.ac.cn

quently fail to capture the long-range dependencies across diverse views, thereby limiting the model's ability to represent complex corneal morphologies.<sup>10,11</sup> These challenges highlight the urgent need for effective multiview data integration that can boost suspect classification without losing discriminative power across all categories.

To tackle these issues, we propose a Transformer-based multiview fusion framework (CMVFT) aimed at enhancing multiclass corneal topography classification—particularly for suspect KCN detection. However, although Transformers have excelled in large-scale image recognition by building global receptive fields via multihead self-attention (MHSA), they lack the inductive bias inherent in CNNs. As a result, Transformers typically require substantial training data and augmentation to generalize well, posing difficulties in small-sample medical imaging scenarios.

In this study, we bridge the strengths of CNN-like local feature extraction and Transformer-based global modeling by introducing a multiscale dynamic module that mitigates the limitations of “input-agnostic” standard convolutions. This design enables a smooth transition from convolutional to Transformer features, allowing us to capture cross-view correlations effectively while adapting to the relatively small KCN dataset. Specifically, the main contributions of our work are as follows:

- Proposed a multiview fusion framework named CMVFT, seamlessly integrating seven standard corneal topography views to address complex multiclass classification scenarios.
- Introduced a Multiscale Attention Module and FusionTransformer, which jointly leverage local fine-grained cues and global morphological dependencies to enhance suspect detection.
- Presented a suspect-oriented classification strategy by incorporating ArcFace and Focal Loss, effectively mitigating feature overlap between suspect and normal classes, and boosting suspect AUC from 79.7% to 86.3%.
- Demonstrated state-of-the-art performance on three-class corneal topography tasks (normal, suspect, KCN), with a 6.6% AUC improvement over mainstream methods, thereby advancing the field of multiview medical image analysis.

The core novelty here is our first-of-its-kind application of a Transformer-driven multiview fusion method specifically tailored for corneal topography, offering a new technical pathway to resolve the suspect-category classification challenge in complex multiclass tasks. Theoretically, this work expands the frontier of deep learning in ophthalmic image analysis by presenting a multiview integration strategy that balances CNN-based inductive bias with Transformer's global capabilities. Practically, it significantly augments the sensitivity and accuracy of clinical diagnoses, especially for early or borderline KCN presentations, thus informing timely interventions and reducing the risk of se-

vere ectatic progression—ultimately improving patient outcomes.

The remainder of this paper is organized as follows: Section 2 surveys the related work on keratoconus classification and multiview fusion. Section 3 details the CMVFT architecture, including the multiscale attention mechanism and the FusionTransformer. Sections 4 and 5 present our experimental setup and results, with emphasis on the suspect class performance. Finally, Section 6 concludes the paper by summarizing the research findings and outlining future directions for multiview medical image analysis.

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## RELATED WORK

Traditional machine learning methods have been widely applied in corneal topography classification, using manually designed features for classification, primarily focusing on extracting parameters from corneal topography or tomographic data. Souza et al<sup>12</sup> combined methods such as Support Vector Machines (SVM), Multilayer Perceptron (MLP), and Radial Basis Function Neural Networks (RBFNN), classifying four categories of eyes (normal, astigmatism, KCN, and postlaser refractive surgery) based on 11 parameters extracted from the Orbscan II device. Their results demonstrated that classification performance was significantly higher than models using single features. However, the model's performance was primarily focused on binary classification tasks (such as normal vs KCN) with limited discussion on four-class classification and did not consider the potential of multiview information fusion. Aatila et al<sup>13</sup> employed Sequential Forward Selection (SFS) to reduce feature dimensions, achieving 98% accuracy for binary classification (normal vs KCN) and 95% accuracy for four-class classification (normal, early-stage KCN, subclinical KCN, and KCN) in a Random Forest classifier. However, this study relied on manual feature selection, ignored potential higher-order information within the data, and might perform poorly when applied to different devices or data distributions, as the training data was limited to parameters from specific devices. Cohen et al<sup>14</sup> trained a Random Forest classifier with 8526 corneal tomography images for a three-class task (normal, suspicious, and KCN), achieving a classification accuracy of 91.5%, sensitivity of 94.7%, and specificity of 89.8%. While this study performed excellently in the three-class task, its features were still based on parameters extracted from specific instruments, failing to fully explore automatic feature extraction or more complex feature combination strategies. Traditional methods rely heavily on manual feature extraction, limiting scalability and adaptability. In fact, this is a labor-intensive and time-consuming process dependent on experts in the field. However, in deep learning, this step is handled automatically by algorithms. Furthermore, machine learning algorithms face performance platform issues,<sup>15</sup> where the per-

formance remains constant even with an increase in data volume. Deep learning algorithms typically perform better with an increased amount of data. In this regard, many researchers have turned to deep learning methods to improve diagnostic efficiency.

In recent years, deep learning methods have gradually become the core technology for corneal topography classification tasks, demonstrating significant advantages due to their automated feature extraction capabilities and ability to handle multiview data. Shi et al<sup>16</sup> combined Scheimpflug and OCT data and used neural networks to perform a three-class classification task for normal, subclinical KCN, and KCN. The model achieved an AUC of 0.93 for subclinical KCN. However, their study primarily focused on feature extraction and combination from a single data modality, failing to fully explore the potential of multimodal data. Additionally, the small data scale (121 images) could impact the model's generalizability and stability. Al-Timemy et al<sup>10</sup> employed transfer learning methods, integrating multiple pretrained models (such as AlexNet and SqueezeNet), and achieved binary classification for normal vs KCN by probabilistic fusion, with the classification accuracy of the single model reaching 94.8%, and the ensemble model's accuracy further improving to 98.3%. However, the multiclassification capability of this study has not been explored, and the final classification results were obtained solely through probabilistic fusion, lacking interpretability regarding whether the features were effectively utilized. Moghaddam et al<sup>17</sup> used Pentacam imaging data, combining posterior surface height data with anterior surface parameters, to develop a multiparameter-based method for corneal severity classification, significantly improving sensitivity and specificity for different stages of KCN. The main limitations of these deep learning methods include the high demand for large-scale labeled data, limited capability in multiclass tasks, and insufficient exploration of model interpretability and multimodal data fusion.

The growing adoption of multiview fusion techniques in medical image analysis has emerged as a prominent research focus, demonstrating enhanced diagnostic accuracy through cross-modal information integration. For instance, Tang et al<sup>18</sup> proposed an instance-wise multiview visual fusion mechanism for zero-shot learning, which aligns multiview representations and enforces complementary principles to improve performance and mitigate domain shift issues. This approach offers a more sophisticated method for multiview fusion compared to traditional feature concatenation. Liu et al<sup>19</sup> introduced MSMV-Net, which performs exceptionally well in multiscale and multiview fusion, surpassing existing methods in the task of lung tumor segmentation. Yang et al<sup>20</sup> proposed a novel multiview feature fusion and density-based oversampling technique to handle imbalanced data, effectively combining various protein sequence features for amyloid protein prediction. This multiview approach leverages the complementary properties of different views to improve prediction performance,

similar to how multiview fusion techniques enhance medical image analysis. Additionally, He et al<sup>21</sup> proposed an adaptive dual-view fusion framework, which integrates T2-weighted Imaging (T2WI) and Apparent Diffusion Coefficient (ADC) sequence data from MRI, achieving significant advantages in preoperative prostate cancer grading. Alsulami et al<sup>22</sup> utilized a multiview multifeature fusion model, MV-MFF, to improve pneumonia classification accuracy by leveraging information from different views of X-ray images. Li et al<sup>23</sup> proposed a multiview feature fusion encoding and memory-driven report generation model, which combines the shallow spatial features and deep semantic features of OCTA images to create detailed visual representations of lesion characteristics and uses a relational memory module to enhance the generalization ability of diagnostic report generation models. These studies demonstrate that multiview fusion can fully exploit the complementary properties between modalities. However, most current multiview fusion methods are custom-developed for specific tasks or datasets, and there is no universal framework for corneal image analysis. Further research on fusion methods suitable for corneal multiview data is of great significance. Unlike the aforementioned studies, this research is the first to propose a Transformer-based multiview fusion method for corneal topography classification. Due to its advantage in modeling long-range dependencies, the Transformer can effectively capture and integrate multiview features, thereby providing a comprehensive representation of corneal morphological information. Additionally, the Transformer architecture is not limited to a specific network structure but provides a flexible framework for the comprehensive use of multimodal data. By incorporating the fusion of seven corneal topography datasets, this method demonstrates significant advantages in complex multiclass tasks, particularly in the classification of suspicious categories, addressing the limitations of existing methods in multiview feature integration and classification performance. This innovative approach offers a new solution for complex tasks in corneal topography analysis and opens up new directions for future research.

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## METHODOLOGY

As shown in [Figure 1](#), the corneal topography classification method CMVFT proposed in this study, based on deep convolutional networks and a multi-input Transformer architecture, aims to classify corneal topographies into three categories: normal, suspect KCN (suspect), and KCN. Each sample consists of seven standard views, including the corneal thickness map, anterior eccentricity map, posterior eccentricity map, anterior elevation map, posterior elevation map, anterior sagittal map, and posterior sagittal map.

The overall network architecture consists of several key modules: feature extractor; Multiscale Attention Module;

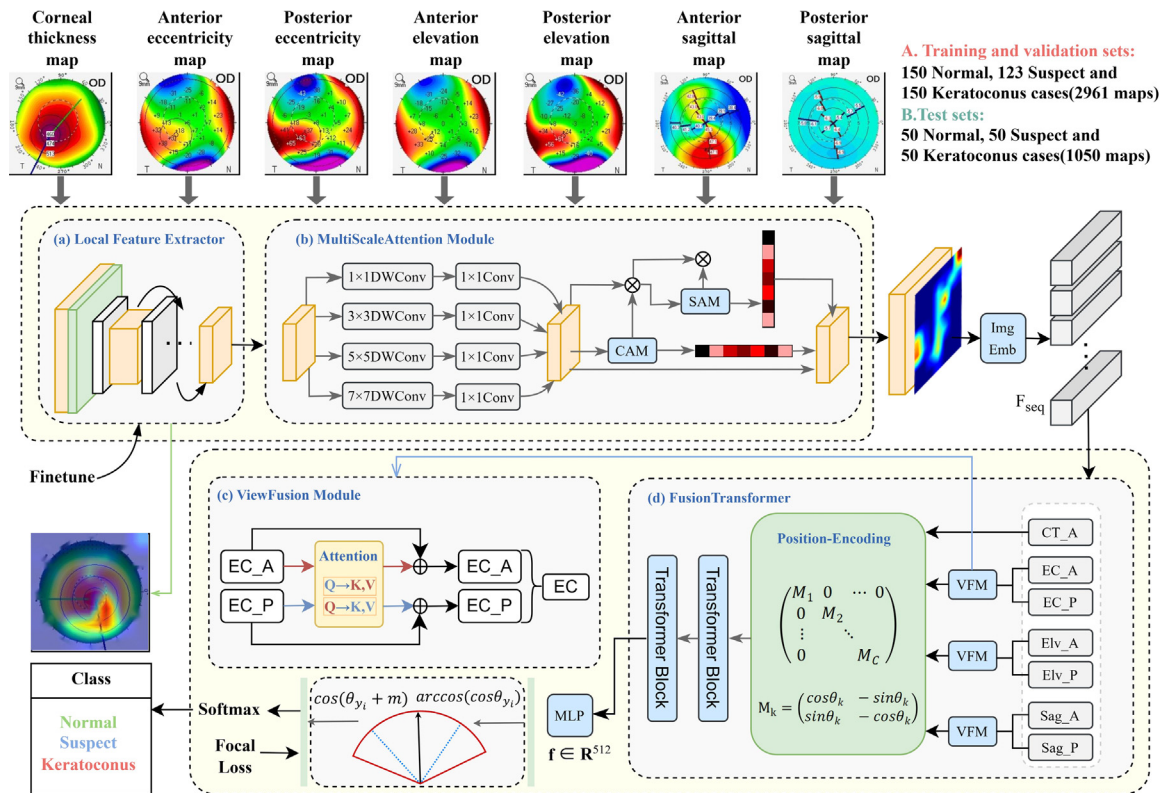


FIGURE 1. The network architecture of CMVFT. CMVFT takes seven corneal topographies as input and extracts multiscale image features through the feature extractor and multiscale attention module. The features are then fused using the FusionTransformer, with additional information fusion applied to specific view pairs.

Multiview Feature Fusion Transformer; and loss function design. First, the fourth layer of the pretrained ResNet-50 is used as the feature extractor, effectively mitigating the overfitting problem caused by insufficient samples through transfer learning. Then, the Multiscale Attention Module is employed to extract multiscale features and capture crucial information. Subsequently, channel and spatial attention mechanisms further enhance the feature representation capability.

The features processed by the attention mechanism are input into the Transformer-based FusionTransformer module, which effectively fuses the features from the seven views. Finally, the feature dimensions are reduced to 512 through a Multilayer Perceptron (MLP), and the ArcFace loss function is used for training and optimizing the three-class classification task to improve classification accuracy and discriminability.

The following section will provide a detailed description of the design and implementation approach for each module.

- **LOCAL FEATURE EXTRACTOR:** The standard ResNet-50 is used as the feature extractor to fully leverage its powerful capabilities in image feature extraction. ResNet-50, by introducing residual modules, effectively alleviates the

gradient vanishing problem in deep networks and is capable of extracting rich hierarchical feature representations.<sup>24</sup> Specifically, this study selects the fourth layer (layer4) of ResNet-50 as the final feature output to obtain a high-dimensional 2048-dimensional feature representation.

To address the issue of limited samples in the corneal topography dataset, this research adopts a fine-tuning strategy for transfer learning. First, the ResNet-50 model pretrained on large-scale image datasets, such as ImageNet, is used as the initial weight. This strategy leverages the learning from the pretrained model on general image features, improving performance on the specific task at hand.<sup>25</sup>

During the fine-tuning process, to prevent overfitting and speed up training, the weights of the first three layers (layer1 to layer3), which are close to the input, are frozen, and only the parameters of layer4 are fine-tuned. This approach not only preserves the low-level features (such as edges and textures) extracted by the earlier layers but also allows higher-level features to be adjusted according to the specific task, thus better adapting to the classification requirements of corneal topography.

- **MULTISCALEATTENTION MODULE:** The MultiScaleAttention Module (MSAM) is designed to provide richer and more hierarchical input features for the subsequent Fusion-



Transformer through multiscale feature fusion and attention reweighting. This design is particularly important for corneal topography classification, as corneal topographies contain structural features at different spatial scales. For instance, local abnormalities may manifest as high-frequency texture changes at smaller scales, while overall curvature and radius information requires a larger receptive field for capturing. In scenarios with limited training data, multiscale representation is especially valuable, as it fully leverages multilevel information to enhance model robustness.

Despite the advantages of multihead self-attention (MHSA) in covering a global receptive field, it lacks the inductive bias of CNNs, resulting in weaker generalization when data are scarce. Although Swin Transformer introduced shifted window attention to embed partial inductive bias and reduce computational cost, its window-based local attention still restricts the effective receptive field. Meanwhile, many hybrid networks (eg, PVT) combine convolution with self-attention to incorporate CNN-like priors, but the use of standard convolution (which is input-agnostic) often limits performance gains. Convolution kernels do not adapt to varying inputs, creating a representation gap between convolutional and attention-based features.

To address these limitations, our MSAM is custom-designed as a multiscale, dynamic feature extraction module. It selectively fuses information across multiple convolutional kernels and attention-based reweighting, thus blending local inductive bias with the flexibility of self-attention. This approach ensures a smooth transition between convolutional and Transformer operations, retaining the strengths of both and facilitating comprehensive representation learning prior to the FusionTransformer stage.

In the medical scenario of corneal topography classification, different pathological types (such as normal, suspect KCN, and KCN) may show significant differences in both small-scale features (such as local irregular points) and larger-scale features (such as overall corneal morphological changes). Therefore, multiscale feature extraction provides a rich feature foundation suitable for integrating multitype information for the subsequent Transformer module. In this study, a feature pyramid structure<sup>26</sup> is used in multi-branch configurations with convolution kernels of sizes  $k \in \{1, 3, 5, 7\}$ . However, directly using standard convolutions would significantly increase computational and parameter costs. To mitigate this, the study introduces depthwise separable convolutions<sup>27</sup> combined with  $1 \times 1$  convolutions in each multiscale branch to reduce computational costs and improve training and inference efficiency. The complexity of traditional convolutions (excluding biases) is approximately:

$$O(C_{in} \times C_{out} \times K^2 \times H \times W) \quad (1)$$

where  $C_{in}$  and  $C_{out}$  are the number of input and output channels respectively,  $K$  is the convolution kernel size, and  $H$  and  $W$  are the feature map size. While depth-separable

convolution decomposes the standard convolution into two parts, Depthwise Convolution and Pointwise Convolution, with a complexity of approx:

$$O(C_{in} \times C_{out} \times K^2 \times H \times W) + O(C_{in} \times C_{out} \times K^2 \times H \times W) \quad (2)$$

Compared to standard convolution, when  $C_{out}$  and  $C_{in}$  are close, this decomposition can significantly reduce the computational cost. Referencing the successful application of depthwise separable convolution in MobileNet,<sup>28</sup> this study uses parallel paths with depthwise convolution kernels of sizes 1, 3, 5, and 7 in the multiscale structure. Through their respective Depthwise Convs and subsequent  $1 \times 1$  Convs, the channel number is reduced from 2048 to 512, with the output of each path being:

$$X'_k = \text{Conv}_{1 \times 1}(\text{DWConv}_{k \times k}(X')), k \in \{1, 3, 5, 7\} \quad (3)$$

where  $X' \in \mathbb{R}^{(B \times V) \times 2048 \times H \times W}$  is the expanded input feature tensor. The four subpath outputs are obtained by splicing the four subpaths on the channel dimension:

$$X_{\text{fuse}} = \text{Concat}(X'_1, X'_3, X'_5, X'_7) \in \mathbb{R}^{(B \times V) \times 2048 \times H \times W} \quad (4)$$

The channel dimensions are restored to match the input, ensuring channel consistency for multiscale information. Subsequently, this study introduces CBAM<sup>29</sup> to apply attention weighting on both the channel and spatial dimensions to  $X_{\text{fuse}}$ :

$$X_{\text{att}} = \text{CBAM}(X_{\text{fuse}}) \quad (5)$$

The resulting  $X_{\text{out}} \in \mathbb{R}^{(B \times V) \times 2048 \times H \times W}$  provides a multiscale feature representation, rich in both local fine-grained features and global structural information, for the subsequent feature fusion in the Transformer. The multiscale attention module offers multiscale, attention-optimized feature representations to the following FusionTransformer module, enabling the Transformer to more effectively model multiview features globally. In the self-attention mechanism of the Transformer, correlations between different views and spatial locations are effectively utilized: the multiscale features ensure that the Transformer has enough information for both fine textures and global structures, thereby facilitating better feature fusion and context information integration.

Preliminary experimental results (which will be discussed in detail in the experimental section) show that, compared to the baseline model without a multiscale module or attention mechanism, the introduction of the Multiscale At-

tention Module significantly improves classification performance and feature robustness. This result demonstrates that the reasonable integration of multiscale and attention mechanisms provides substantial assistance in the multi-view corneal image classification task.

- **VIEWFUSION MODULE:** The ViewFusion Module implements Multihead Attention between two views by deriving the query (Q) matrix from the source view A, and the key (K) and value (V) matrices from the target view P. Since the multihead attention layer is asymmetric, the result depends on the choice of source and target views. To address this, the module computes multihead attention in both directions, alternating between the source and target views, ensuring bidirectional fusion. Let the source view A and target view P have their feature representations denoted as A and P. The multihead attention computation process is as follows:

$$\text{Attention} = (Q, K, V) = \text{softmax}\left(\frac{Q_A K_P^T}{\sqrt{d_k}}\right) V_P \quad (6)$$

where  $Q_A = W_Q A$ ,  $K_P = W_K P$ ,  $V_P = W_V P$ , and  $W_Q$ ,  $W_K$ ,  $W_V$  are the learned weight matrices and  $d_k$  is the dimension of the key vector.

To achieve bidirectional attention fusion, the module first calculates the attention output  $O_{P \rightarrow A}$  with View A as the source and View P as the target, then calculates the attention output  $O_{A \rightarrow P}$  with View P as the source and View A as the target. Finally, these two outputs are added to the original features of View A and View P, respectively:

$$A' = A + O_{P \rightarrow A} \quad (7)$$

$$P' = P + O_{A \rightarrow P} \quad (8)$$

This bidirectional attention mechanism ensures sufficient exchange and fusion of information between the source and target views, thus enhancing the representational ability of the features. The module is applied to each pair of the same-type views (eg, EC-A and EC-P, Elv-A and Elv-P, Sag-A and Sag-P), ultimately generating the fused feature representations.

- **FUSIONTRANSFORMER:** The FusionTransformer module is designed to efficiently fuse features from seven different perspectives to achieve accurate corneal topography classification. Considering the quadratic complexity  $O(N^2)$  of the standard Transformer architecture<sup>30</sup> when handling long sequences, directly inputting the features from all seven perspectives into the Transformer would lead to significant computational resource consumption. To address this, this study designs the ViewFusion Module, which

first fuses features from the same type of perspective, significantly reducing the length of the one-dimensional sequence required for subsequent feature fusion, thereby optimizing computational efficiency.

The FusionTransformer adopts a basic two-layer Transformer architecture, where each layer includes a multihead self-attention mechanism and a feed-forward neural network. The input features are first preprocessed by the ViewFusion Module, which fuses features from the same type of perspectives to generate shorter sequences for processing by the Transformer. Specifically, the ViewFusion Module applies a multihead attention mechanism to each pair of same-type views (such as EC-A and EC-P, Elv-A and Elv-P, Sag-A and Sag-P), calculating attention in both directions to achieve bidirectional feature fusion.

#### Feature fusion and transformer processing

After processing with the ViewFusion Module, the feature sequence length is reduced from the original seven perspectives to the fusion features of three pairs of views. These fused features are then input into a two-layer basic Transformer architecture for further feature fusion and global context modeling. The self-attention mechanism of the Transformer captures the correlations and global dependencies between different perspectives, thereby enhancing the overall expressive power of the features.

Specifically, each layer of the FusionTransformer includes multihead self-attention and a feed-forward neural network, as shown in the following formula:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (9)$$

$$\text{head}_i = \text{Attention}(Q W_i^Q, K W_i^K, V W_i^V) \quad (10)$$

where  $W_i^Q$ ,  $W_i^K$ ,  $W_i^V$ ,  $W^O$  are learned weight matrices, and  $h$  represents the number of heads. To enhance the representation of positional information, this study incorporates Rotary Position Embedding (RoPE)<sup>31</sup> in the Transformer. RoPE achieves unbiased positional encoding by rotating the position encoding with the query and key vectors, while preserving the relative positional information. Specifically, RoPE defines a rotation matrix  $\text{Rot}(p)$  for each position  $p$ , which takes the form of:

$$\text{Rot}(p) = \begin{bmatrix} \cos(\theta_p) & -\sin(\theta_p) \\ \sin(\theta_p) & \cos(\theta_p) \end{bmatrix} \quad (11)$$

where  $\theta_p$  is the angle encoding associated with position  $p$ , typically linked to frequency. RoPE is then applied to the query and key vectors as follows:

$$Q_p = Q_p \cdot \text{Rot}(p) \quad (12)$$

$$K_p = K_p \cdot \text{Rot}(p) \quad (13)$$

This rotational position embedding method not only preserves relative positional relationships but also introduces positional information into the self-attention computation, allowing the Transformer to better capture spatial location information when processing multiview features.

#### Classification head

The feature dimension output by FusionTransformer is 2048. To meet the requirements of the ArcFace loss function,<sup>32</sup> global average pooling (Global Average Pooling) is used to reduce the feature dimension from 2048 to 512:

$$F_{\text{pool}} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_{\text{out}}(i, j) \quad (14)$$

Subsequently, a multilayer perceptron (MLP) is used to further map the pooled features into a 512-dimensional space, and the ArcFace loss function is applied for the training and optimization of the three-class classification task. ArcFace enhances the discriminative power between categories by calculating the cosine angle, as shown in the following formula:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s \cos \theta_j}} \quad (15)$$

where  $s$  is the scaling factor,  $m$  is the angular margin,  $\theta_{y_i}$  is the cosine similarity between sample  $i$  and its correct class  $y_i$ , and  $\theta_j$  is the cosine similarity between sample  $i$  and other classes  $j$ .

In this study, to enhance the model's performance in the corneal topography three-class classification task, particularly in classifying the "Suspect" category, we adopted a combined optimization strategy using both ArcFace Loss and Focal Loss.<sup>33</sup> ArcFace Loss strengthens the discriminative power between categories, while Focal Loss alleviates the class imbalance issue. The combined loss function is defined as follows:

$$L_{\text{Total}} = \lambda_1 L_{\text{ArcFace}} + \lambda_2 L_{\text{Focal}} \quad (16)$$

Where the Focal Loss is defined as:

$$L_{\text{Focal}} = -\frac{1}{N} \sum_{i=1}^N \alpha (1 - p_t)^\gamma \log(p_t) \quad (17)$$

Here,  $p_t$  is the model's predicted probability that sample  $i$  belongs to its true class,  $\alpha$  is the class weight, and  $\gamma$  is the focusing parameter used to adjust the contribution of easily classified samples to the loss.

Compared to traditional cross-entropy loss, ArcFace Loss introduces an angular margin to enhance feature discriminability by achieving tighter intraclass aggregation and larger interclass separation. Focal Loss, on the other hand, reduces the weight of easily classified samples, allowing the model to focus more on the minority and hard-to-classify samples. This combination significantly improves classification performance, particularly in cases with complex class distributions and class imbalances.

## DATASET

- **ORIGINAL DATASET:** This study uses a publicly available corneal image dataset,<sup>11</sup> which includes data from 573 eyes. The dataset consists of 423 eyes in the original training and validation set (150 normal (NOR) eyes, 150 KCN (Keratoconus, KCN) eyes, and 123 suspect KCN (SUSPECT) eyes) and 150 eyes in the test set (50 NOR eyes, 50 KCN eyes, and 50 SUSPECT eyes). For the purpose of conducting five-fold cross-validation and three-way partition experiments, the training and testing sets were combined, as shown in Figure 2. In all partition strategies, stratified sampling was employed to maintain the proportion of each category consistent across different splits, with a random seed of 42 used to ensure the reproducibility of results, while ensuring no overlapping samples between partitions.

The criteria for diagnosing suspect KCN are based on relevant clinical guidelines and studies,<sup>34,35</sup> including abnormal local thickening or asymmetric butterfly patterns shown on the corneal topography, corneal curvature exceeding 47.00 diopters (D), astigmatism greater than 1.50 D, central corneal thickness below 500 micrometers, and abnormalities in the anterior surface elevation, progression thickness increase (PTI), or the Belin–Ambrosio Ectasia Display (BAD-D) index in elevation map analysis. All images underwent standardization before use, including deidentification, intensity normalization, and alignment, to ensure consistency across different imaging devices and sessions. The training dataset includes images captured using different Pentacam instruments and settings, enhancing the model's robustness to instrument variability.

- **DATASET PREPROCESSING AND DATA AUGMENTATION:** In this study, the first step involved preprocessing the corneal images by detecting and repairing text to eliminate any textual information that could interfere with the classification model. Specifically, the method used keras-ocr to detect text regions in the images, and the OpenCV inpaint function was applied to repair these regions, ensuring the images were clean and consistent.<sup>36</sup> Subsequently, to improve the model's generalization ability and robustness, several data augmentation techniques were applied to the training dataset, including resizing the images to  $224 \times 224$  pixels, random horizontal flipping (50% prob-

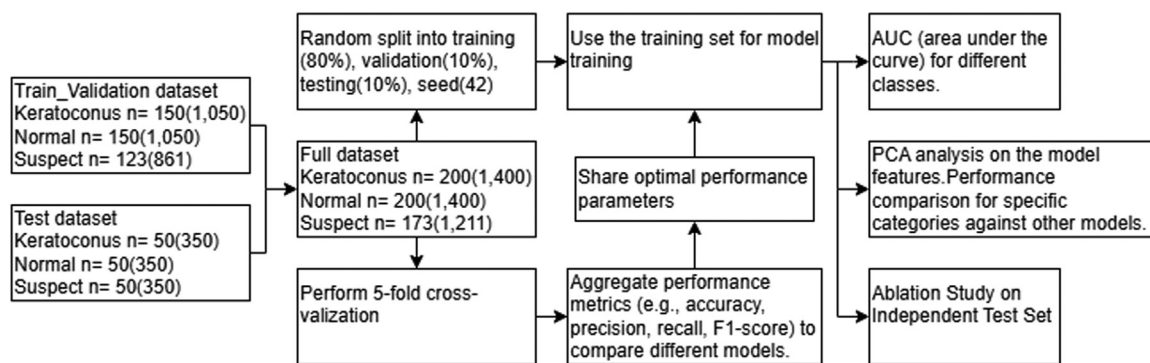


FIGURE 2. Dataset processing and model training flowchart.

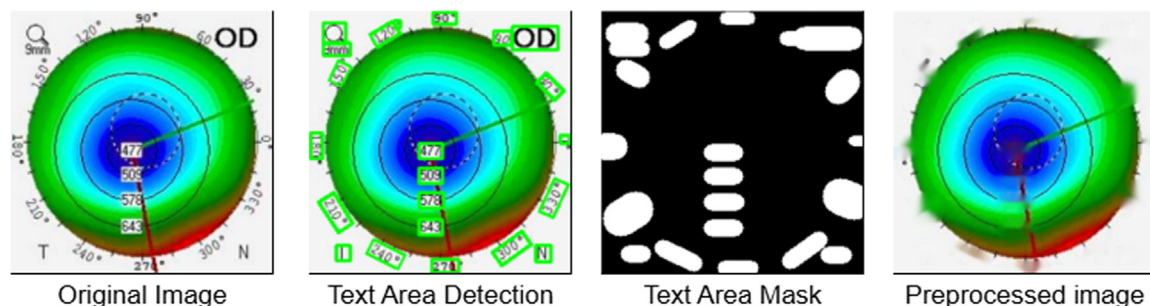


FIGURE 3. Preprocessing and denoising flowchart.

ability), slight rotation (up to 5 degrees), random gamma correction ( $\gamma=1.5$ , 50% probability), edge enhancement (30% probability), contrast jitter ( $\pm 20\%$ , 50% probability), Gaussian blur (kernel size  $5 \times 5$ ,  $\sigma$  range 0.1-2.0, 30% probability), and random scaling and cropping to maintain image size at  $224 \times 224$  pixels. Finally, the images were converted into tensors and normalized (mean: [0.485, 0.456, 0.406], SD: [0.229, 0.224, 0.225]). The preprocessing method is illustrated in Figure 3. All experiments in this study were conducted on the deidentified dataset.

## EXPERIMENT

The experimental section of this study systematically evaluates the proposed corneal topography classification model CMVFT, particularly focusing on the advantages of the Transformer-based fusion method, FusionTransformer, in handling multiview data, and validating its effectiveness in the corneal topography classification task. Specifically, the experiment aims to conduct a comprehensive comparison with existing fusion methods and classical deep learning architectures to reveal the superiority of the proposed method in terms of classification accuracy, computational efficiency, and computational complexity. Existing fusion methods are typically categorized into decision-level fusion and feature-level fusion. Decision-level fusion methods (eg, weighted voting and majority voting<sup>10</sup>) often combine

the prediction results after processing each view independently. While simple to implement, they overlook the complex relationships between views and fail to fully exploit the complementary information in multiview data, which is highly disadvantageous in corneal tasks where data is limited. In contrast, feature-level fusion (eg, weighted averaging, concatenation, etc.) directly merges features in the feature space, integrating multiview feature information. However, it still suffers from high computational complexity and the challenge of modeling dependencies between views. In this study, the FusionTransformer is introduced to precisely model the relationships between features from different views, using its two-layer Transformer architecture and multihead self-attention mechanism. Combined with the multihead attention bidirectional fusion mechanism in the ViewFusion Module, this study can more efficiently capture the complex interactions and global contextual information of multiview features. A comparison with traditional fusion methods verifies the advantages of the proposed method in multiview feature fusion.

We will also compare it with traditional Convolutional Neural Network (CNN) architectures, including classic deep convolutional models such as ResNet, VGG, and lightweight CNN models (eg, MobileNet, EfficientNet, etc.). These deep learning architectures perform excellently in single-view image classification tasks but lack effective feature fusion mechanisms in handling multiview data, thus failing to fully utilize the complementary information between different views. This experiment compares Fusion-



Transformer with these classic CNN models to showcase its advantages in corneal topography classification, particularly in the fusion of features across views and its contribution to improved classification accuracy. Finally, this experiment will demonstrate the unique advantages of FusionTransformer in multiview feature fusion, especially its ability to integrate multiview information through self-attention mechanisms. By comparing the proposed method with existing fusion methods and mainstream deep learning models, this study aims to provide empirical support for the application of CMVFT in medical image classification tasks and demonstrate its innovation and practicality in handling multiview data.

• **EXPERIMENTAL SETUP:** The experiments were conducted using PyTorch 1.12 as the primary deep learning framework, on a workstation equipped with an NVIDIA RTX 4090 GPU (24GB VRAM)  $\times$  1, an 18 vCPU AMD EPYC 9754 128-Core Processor, and 60GB of RAM.

During the training process, the AdamW optimizer was used with a weight decay of  $1 \times 10^{-2}$ . At the beginning of the training, a learning rate warm-up strategy was applied, where the learning rate gradually increased to the initial learning rate of  $1 \times 10^{-5}$  over the first 5 epochs. The learning rate schedule used was WarmupCosineAnnealingLR, which implements a cosine annealing strategy to gradually reduce the learning rate to its minimum value after the warm-up phase, with the minimum learning rate set to 0. The loss weights  $\lambda_{\text{arcface}}$  and  $\lambda_{\text{focal}}$  were both set to 1.0.

• **NETWORK PARAMETER SETTINGS:** In this study, a slightly modified version of ResNet-50 was used as the backbone network to extract features. During the training of ResNet-50, Layer 1, Layer 2, and Layer 3 were frozen, and only Layer 4 was trained. In the MultiScaleAttentionBlock, the output channel number for each branch was set to 512, and the convolution kernel sizes used were  $1 \times 1$ ,  $3 \times 3$ ,  $5 \times 5$ , and  $7 \times 7$ . The features from each branch were processed using depthwise separable convolutions with a stride of 1 and  $1 \times 1$  convolutions, and the final output features were concatenated into a [batch\_size, num\_views, 2048, 7, 7] format. In the FusionTransformer module, the number of Transformer layers was set to 2, with each layer having 8 attention heads, an embedding dimension of 2048, a dropout rate of 0.1, and a DropPath regularization probability of 0.15. The input feature dimension for the ArcFace module was set to 512, with a cosine margin  $m = 0.5$  and a scaling factor  $s = 30.0$ . Focal Loss was used as the loss function, with the focusing parameter  $\gamma = 2.0$ , class balancing parameter  $\alpha = 0.25$ , and loss reduction method set to "mean."

• **EXPERIMENTAL EVALUATION AND METRIC SETUP:** In this experiment, to perform a horizontal comparison of performance with other baseline models and methods, we

adopted a five-fold cross-validation data splitting method. Specifically, the dataset was fixed into five subsets, with one subset selected as the validation set and the remaining four subsets used as the training set for each round. This process was repeated for five rounds, and the model performance was ultimately evaluated based on the average results from these five rounds of training. This method helps increase the stability of the evaluation results and reduces the bias caused by the randomness of data splitting. However, during the experiment, we observed that, due to the limitations of the dataset, five-fold cross-validation may cause overfitting in some folds. Specifically, due to the scarcity of certain features in the dataset, under certain hyperparameter combinations, the feature distribution of a fold may become imbalanced, leading to overfitting on the validation set. To address this issue, when validating the effectiveness of the model's modules, we selected the hyperparameters that yielded the best results in five-fold cross-validation and performed a three-way partition of the dataset, where 80% of the data was used for training, 10% for validation, and 10% for testing.

To comprehensively cf the model's performance in the classification task, we adopted five commonly used evaluation metrics: Precision, Accuracy, Recall, AUC, and F1-Score.

Precision is used to measure the proportion of actual samples belonging to a particular class among all the samples predicted as belonging to that class. The precision calculation formula is:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (18)$$

where TP (True Positive) is the number of samples correctly predicted to be in the positive category and FP (False Positive) is the number of negative samples incorrectly predicted to be in the positive category.

Accuracy is used as a measure of the percentage of correct predictions made by a model across all samples. Accuracy is calculated by the formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (19)$$

Recall is used as a measure of the proportion of all samples that actually belong to a category that the model correctly predicts as being in that category. Recall is calculated by the formula:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (20)$$

AUC is used to evaluate the performance of the model under different classification thresholds, which is particularly suitable for binary classification problems, but can be extended to multiclassification problems as well. The closer

**TABLE 1.** Performance Comparison of Different Deep Learning Models in Corneal Image Classification Tasks

| Model           | AUC                     | Precision               | Recall                  | F1                      |
|-----------------|-------------------------|-------------------------|-------------------------|-------------------------|
| MobileNetV3     | 85.1 (82.8-88.1)        | 68.0 (63.3-72.9)        | 59.9 (51.1-68.2)        | 58.9 (47.1-68.5)        |
| EfficientNet-B0 | 85.6 (82.8-89.8)        | 68.4 (64.5-73.6)        | 57.6 (48.0-70.4)        | 56.5 (43.0-71.0)        |
| EfficientNet-B3 | 86.4 (81.0-91.4)        | 70.4 (66.9-76.6)        | 68.8 (64.3-74.2)        | 68.2 (63.3-73.9)        |
| VGG16           | 90.6 (87.5-93.9)        | 77.3 (74.0-82.2)        | 75.9 (71.0-81.9)        | 76.1 (71.6-81.9)        |
| ResNet-50       | 89.3 (86.9-91.4)        | 75.2 (70.1-80.8)        | 74.9 (69.7-79.3)        | 74.7 (69.6-78.8)        |
| ResNet-152      | <b>91.8 (90.8-92.7)</b> | 79.7 (78.2-82.0)        | 78.4 (76.8-81.1)        | <b>78.6 (76.9-81.2)</b> |
| Ours            | 90.9 (87.9-92.3)        | <b>80.6 (77.2-83.8)</b> | <b>79.4 (76.0-80.9)</b> | 78.5 (73.4-80.7)        |

Bold values indicate the best performing results for each evaluation metric across compared models.

the AUC value is to 1, the better the model's classification ability is. The computation of the AUC value is usually obtained by plotting the ROC curve, where the relationship between False Positive Rate (FPR) and True Positive Rate (TPR) is as follows. FPR and True Positive Rate (TPR) are related as follows:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (21)$$

The AUC value, on the other hand, is the area under the ROC curve, which is used to evaluate the performance of a multiclassification model by calculating the AUC for each category and averaging it.

The F1 score (F1-Score) is the reconciled average of precision and recall, which combines the advantages and disadvantages of precision and recall to provide a more comprehensive assessment of the model performance. The F1 score is calculated by the formula:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

To aid clinical translation, we additionally report the expected positive predictive value (PPV) under real-world prevalence conditions. PPV is estimated based on the following standard epidemiological formula:

$$\text{PPV} = \frac{\text{Sensitivity} \times \text{Prevalence}}{\text{Sensitivity} \times \text{Prevalence} + (1 - \text{Specificity}) \times (1 - \text{Prevalence})} \quad (23)$$

This allows us to project the reliability of our model predictions for Keratoconus (KCN) and Suspect cases in population-level scenarios.

• **COMPARATIVE EXPERIMENTS:** This study systematically compared the performance of the CMVFT framework with

several classic deep learning models, as shown in Table 1, including MobileNetV3,<sup>37</sup> EfficientNet-B0, EfficientNet-B3,<sup>38</sup> VGG16,<sup>39</sup> ResNet-50, and ResNet-152.<sup>24</sup> Among them, EfficientNet-B0 was used as the main framework in.<sup>11</sup> However, this study did not employ a five-fold cross-validation strategy, which could not effectively mitigate the overfitting risk in specific training-test data splits, potentially limiting the model's generalization ability. In contrast, in,<sup>40</sup> ResNet-50, ResNet-152, and VGG16 were demonstrated to perform excellently in corneal topography analysis tasks, with ResNet-152 achieving the best results. The experiments in this study further confirmed this conclusion, as ResNet-152 achieved the highest AUC value (91.8%) in the comparative experiments. Additionally, we also experimented with ViT<sup>41</sup> and its variant architectures (including ViT-L/32, Swin-T,<sup>42</sup> DeiT,<sup>43</sup> CrossViT,<sup>44</sup> and TNT<sup>45</sup>). However, due to the strong dependence of the ViT architecture on large-scale data, significant overfitting occurred in the corneal task with a limited sample size. Therefore, these experimental results are not presented in this paper.

The experimental results indicate that the CMVFT framework demonstrates superior performance and significant balance across multiple key metrics. With ResNet-50 as the backbone network, CMVFT achieved outstanding results in AUC (90.9%), Precision (80.6%), Recall (79.4%), and F1 score (78.5%). Compared to ResNet-50, CMVFT improved Precision and Recall by 5.4% and 4.5%, respectively. This significant improvement suggests that CMVFT not only identifies positive samples more accurately but also effectively reduces false negatives and false positives, highlighting its advantages in complex classification tasks. In comparison to ResNet-152, although CMVFT slightly lags behind in AUC (90.9% vs 91.8%), it shows a slight advantage in Precision (80.6% vs 79.7%) and Recall (79.4% vs 78.4%). This indicates that CMVFT achieves a better balance between performance and efficiency. It is noteworthy that ResNet-152, being a deeper network, did not exhibit a significant performance advantage in corneal classification tasks, possibly due to parameter redundancy introduced by its complexity. In contrast, CMVFT success-

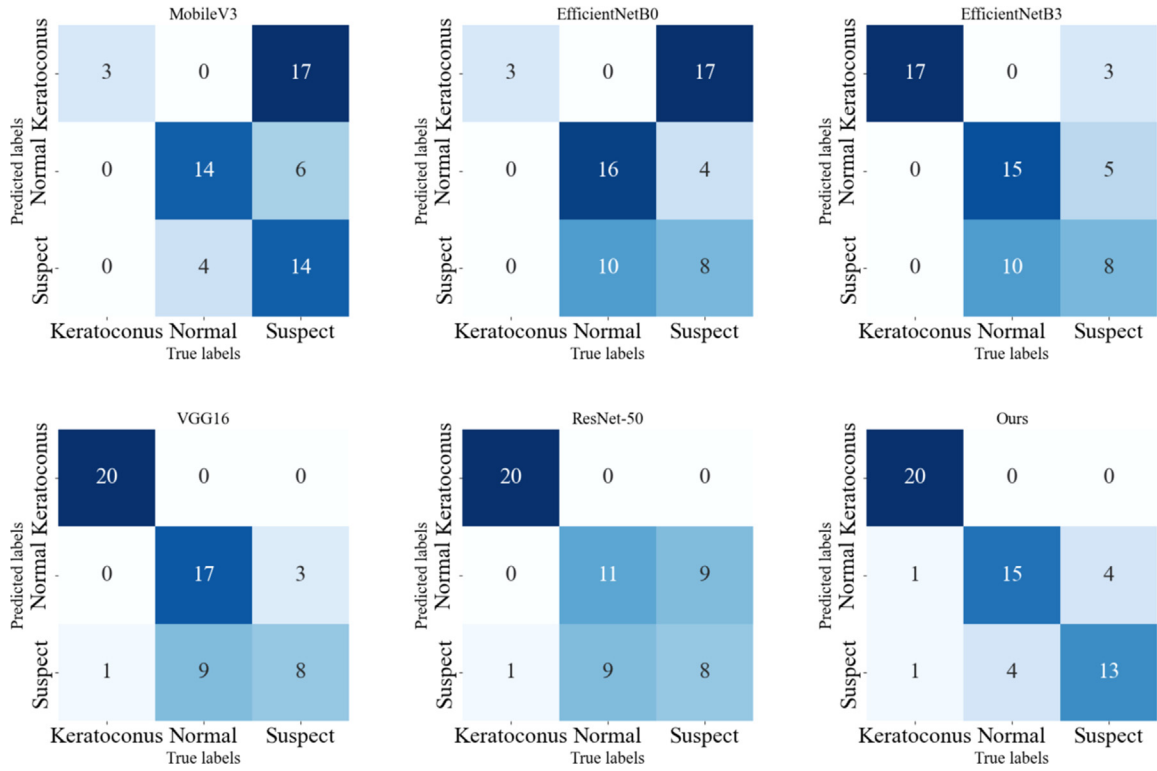


FIGURE 4. Comparison of confusion matrices for different models in corneal KCN detection task.

fully optimized the feature extraction capability of ResNet-50, achieving performance comparable to, or even surpassing, that of ResNet-152 at a lower model complexity.

From the confusion matrix in Figure 4, the performance of different models in the three-class classification task can be intuitively observed. Traditional models (such as MobileV3, EfficientNet, and ResNet-50) tend to optimize overall performance by increasing the classification accuracy of easily classified samples (eg, the Normal class). While this strategy improves the positive sample distribution in the Normal class, it demonstrates some inertia in classifying difficult samples (eg, the Suspect class), resulting in insufficient discriminative ability for these complex samples. In contrast, the multiview fusion architecture exhibits a more comprehensive classification ability. It not only maintains a high accuracy in the Normal class but also shows significant improvement in the recognition of the Suspect class. This indicates that the model is more effective in handling difficult-to-classify samples and avoids the limitation of relying solely on simple samples to boost overall performance.

We evaluated the differences in the performance of various models in classifying difficult-to-classify samples by plotting the ROC curves for the “Suspect” category using six models (Figure 5). The ROC curves were based on a one-vs-all strategy, with the “Suspect” category as the positive class and other categories as negative. The curves show the trade-off between sensitivity (True Positive Rate, TPR)

and specificity (False Positive Rate, FPR) at different decision thresholds. It is important to note that some models’ ROC curves do not start from the (0,0) point, which can be attributed to a combination of factors. In the corneal topography three-class task, the feature space of the “Suspect” category overlaps with that of the “Normal” and “KCN” categories to some extent, as shown in Figure 6. This results in some “Suspect” samples being correctly identified even at very high decision thresholds. Nevertheless, the AUC value of the ROC curve remains a key metric for evaluating classification performance. As can be seen from the analysis of the graph, the CMVFT architecture demonstrates a significant performance advantage in classification (Suspect AUC = 86.3 vs 79.7,  $P = .002$ ). Its curve area (AUC value) is higher than that of other models, reflecting a stronger ability to identify the “Suspect” category.

To further illustrate the impact of the CMVFT architecture on classification tasks, Figure 7 presents the performance of all models in classification. For the KCN class, multiple models exhibit good performance, thanks to the strong feature distribution of the Normal class. However, the CMVFT architecture outperforms all other models in all three-class classification tasks, particularly in the challenging Normal and Suspect categories. The CMVFT model shows a significantly higher AUC compared to other models, demonstrating its exceptional discriminative ability. Further insights can be gained from the statistical significance tests in Table 2, where  $P$ -values for both metrics

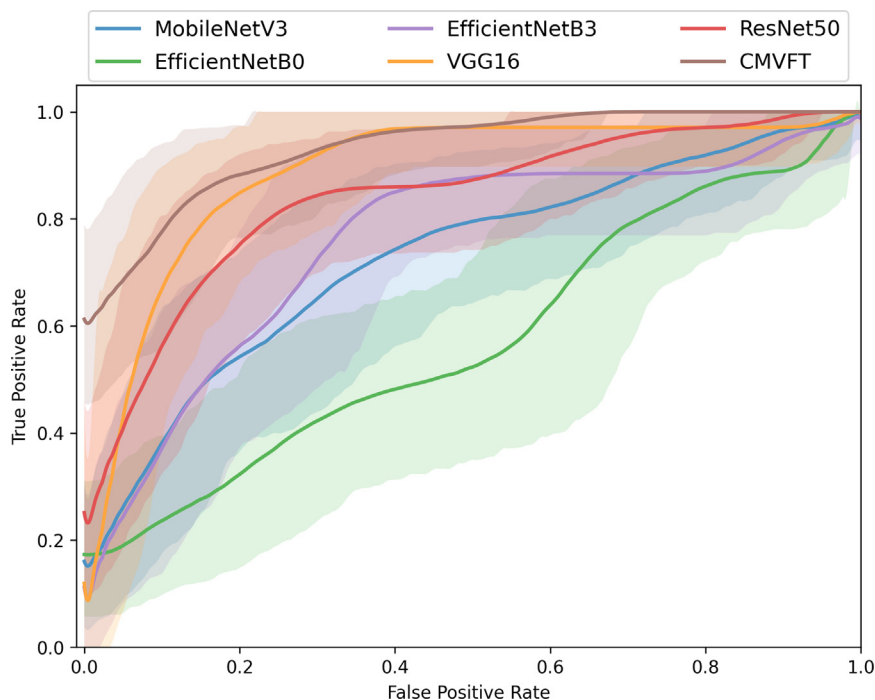


FIGURE 5. ROC curves for the “suspect” category with 95% CIs calculated via bootstrapping. All ROC curves are based on the one-vs-all strategy for the “suspect” category.

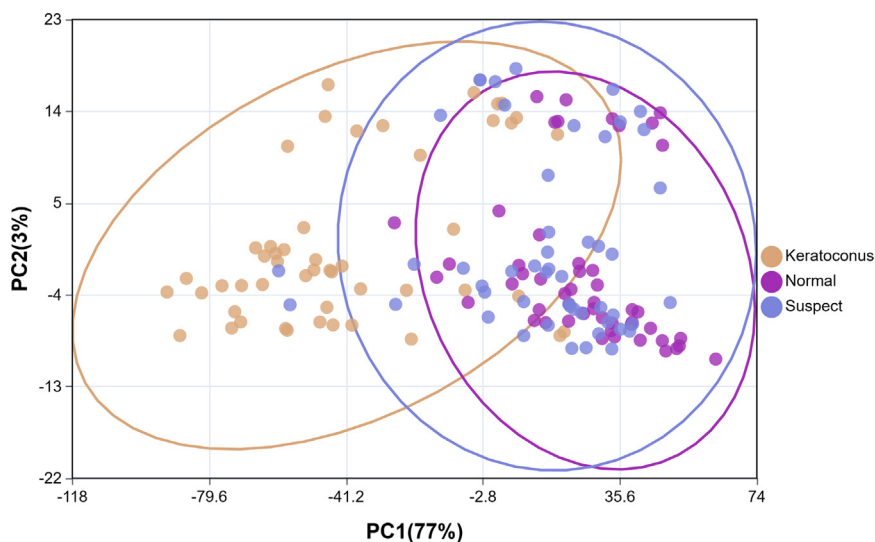


FIGURE 6. Principal component analysis (PCA) plot of corneal topography data. PC1 (77% variance) and PC2 (3% variance) with 95% confidence ellipses for each group.

are significant (all less than 0.05), indicating the statistical significance of the classification results. Notably, for the Suspect\_AUC metric, Cohen’s d values range from 1.98 to 5.18, demonstrating a strong effect size of these models when handling “Suspect” samples, and highlighting their superior performance in identifying difficult-to-classify samples. In the case of a significant feature overlap in the sample distribution, the advantage of the CMVFT architecture becomes even more prominent, as it can more effectively dis-

tinguish challenging samples and enhance the overall classification performance. This improvement reflects not only the impact of model structure and algorithm optimization but also further validates its generalization ability in complex feature spaces.

• **COMPARISON WITH DIFFERENT FUSION STRATEGIES:** In this study, we systematically analyzed the impact of different fusion methods on the performance of the corneal



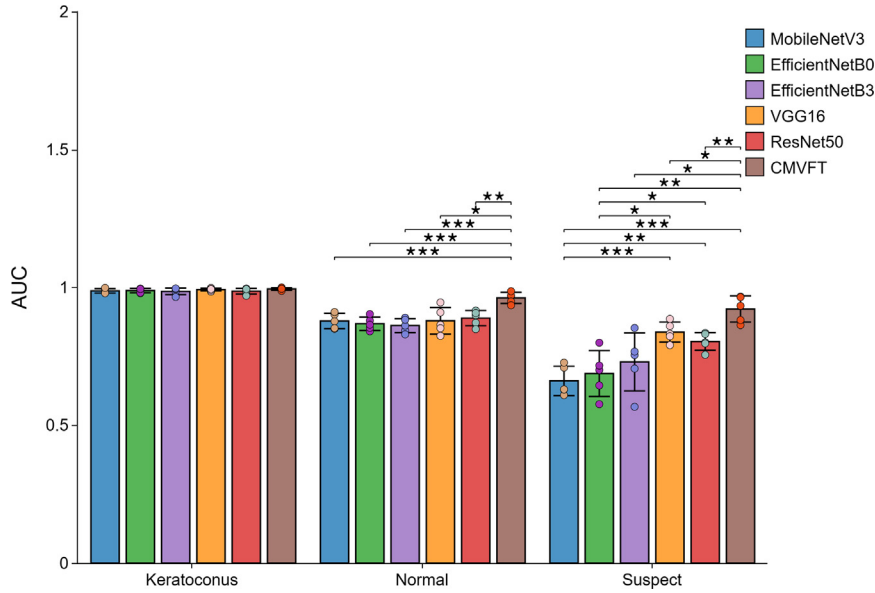


FIGURE 7. Comparison of AUC performance and statistical significance analysis for multimodel classification in keratoconus, normal, and suspect categories.

**TABLE 2.** *P*-Values and Cohen's d Effect Sizes for Different Models in Average\_AUC and Suspect\_AUC Metrics

| Model          | Metric      | <i>P</i> -value | Cohens_d |
|----------------|-------------|-----------------|----------|
| MobileNetV3    | Average_AUC | .000423         | 4.793517 |
| MobileNetV3    | Suspect_AUC | .000496         | 5.175791 |
| EfficientNetB0 | Average_AUC | .000985         | 4.279105 |
| EfficientNetB0 | Suspect_AUC | .003412         | 3.460689 |
| EfficientNetB3 | Average_AUC | .003141         | 2.982429 |
| EfficientNetB3 | Suspect_AUC | .010366         | 2.357123 |
| VGG16          | Average_AUC | .031746         | 2.255117 |
| VGG16          | Suspect_AUC | .049197         | 1.980215 |
| ResNet-50      | Average_AUC | .001336         | 3.232480 |
| ResNet-50      | Suspect_AUC | .002200         | 2.912536 |

Bold values indicate the best performing results for each evaluation metric across compared models.

topography three-class classification task. Tables 3 and 4 show the performance comparison of decision-level fusion and feature-level fusion strategies. The decision-level fusion methods include Majority Voting, Weighted Voting, and Averaging Decision, while feature-level fusion methods cover Concatenation, Weighted Average, and Average Pooling. Decision-level fusion outperforms feature-level fusion in Precision (79.7% vs 75.2%,  $P < .05$ ), Recall (79.3% vs 74.9%,  $P < .05$ ), and F1 score (79.2% vs 74.7%,  $P < .05$ ), with the Weighted Voting method achieving the best results. Furthermore, the line chart in Figure 8 visually demonstrates the performance advantages of decision-level fusion across overall performance metrics.

However, as shown in the confusion matrix analysis in Figure 9, despite decision-level fusion demonstrating supe-

rior overall performance, it exhibits misclassification in the critical KCN category, with some samples being incorrectly classified as Normal or Suspect. This issue may stem from decision-level fusion methods failing to fully utilize the feature information from different views, resulting in insufficient understanding of important features for the KCN category. Considering that the primary goal of corneal topography classification tasks is the early detection of KCN for timely intervention and treatment, misclassification of KCN samples has a more severe negative impact in clinical applications. Therefore, the authors prefer to use the weighted average feature-level fusion method, which, although slightly inferior in overall performance to decision-level fusion, ensures accurate classification of KCN samples, improves the model's interpretability, and enhances the reliability of clinical diagnosis, preventing misclassification of critical categories in pursuit of a falsely high overall accuracy.

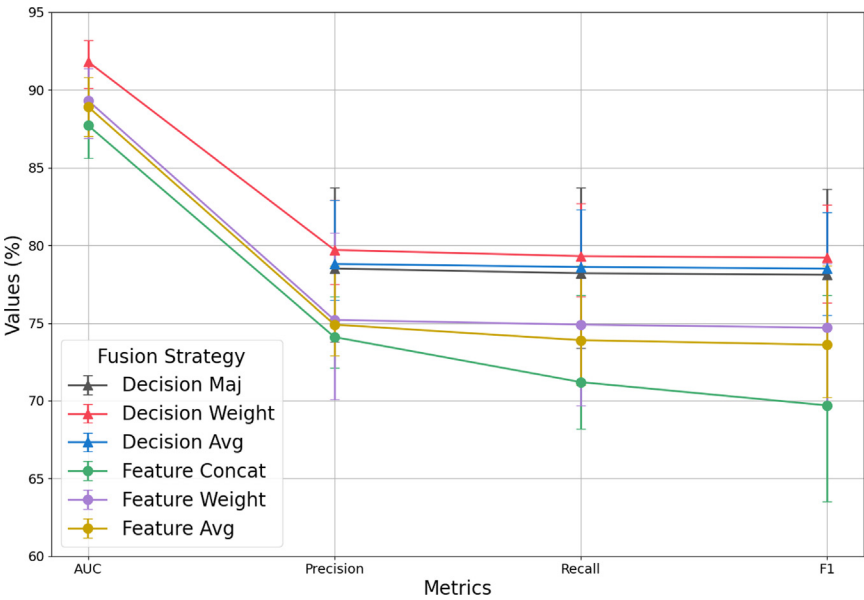
- **VISUALIZATION ANALYSIS:** To ensure the model's ability to recognize the lesion areas, this study used Grad-CAM visualized image samples. As shown in Figure 10, the model can accurately capture the core lesion region in KCN corneal topography images. As the network depth increases, the activated areas gradually focus on the critical central region of the cornea. As shown in Figure 10, A, under the CT-A mode, deep models like VGG16 and CMVFT precisely locate the apex of the cone in KCN. From Figure 10, B through D, it can be seen that under the EC-P and Elv-P modes, CMVFT exhibits exceptional ability, not only identifying sparse areas and asymmetric distributions but also clearly outlining the lesion boundaries and morphological

**TABLE 3.** Performance Comparison of Corneal Topography Classification Using Different Decision-Level Fusion Strategies

| Model                                                                                               | Combination |        |     | Disease Classification |                         |                         |                         |
|-----------------------------------------------------------------------------------------------------|-------------|--------|-----|------------------------|-------------------------|-------------------------|-------------------------|
|                                                                                                     | Maj         | Weight | Avg | AUC                    | Precision               | Recall                  | F1                      |
| ResNet-50                                                                                           | ✓           |        |     | N/A                    | 78.5 (73.8-83.7)        | 78.2 (73.4-83.7)        | 78.1 (73.6-83.6)        |
|                                                                                                     |             | ✓      |     | 91.8 (90.1-93.2)       | <b>79.7 (77.5-82.9)</b> | <b>79.3 (76.7-82.7)</b> | <b>79.2 (76.3-82.6)</b> |
|                                                                                                     |             |        | ✓   | N/A                    | 78.8 (76.5-82.9)        | 78.6 (74.9-82.3)        | 78.5 (75.5-82.1)        |
| Bold values indicate the best performing results for each evaluation metric across compared models. |             |        |     |                        |                         |                         |                         |

**TABLE 4.** Performance Comparison of Corneal Topography Classification Using Different Feature-Level Fusion Strategies

| Model                                                                                               | Combination |        |     | Disease Classification  |                         |                         |                         |
|-----------------------------------------------------------------------------------------------------|-------------|--------|-----|-------------------------|-------------------------|-------------------------|-------------------------|
|                                                                                                     | Concat      | Weight | Avg | AUC                     | Precision               | Recall                  | F1                      |
| ResNet-50                                                                                           | ✓           |        |     | 87.7 (85.6-89.3)        | 74.1 (72.1-76.7)        | 71.2 (68.2-76.8)        | 69.7 (63.5-76.8)        |
|                                                                                                     |             | ✓      |     | <b>89.3 (86.9-91.4)</b> | <b>75.2 (70.1-80.8)</b> | <b>74.9 (69.7-79.3)</b> | <b>74.7 (69.6-78.8)</b> |
|                                                                                                     |             |        | ✓   | 88.9 (87.0-90.8)        | 74.9 (72.9-78.7)        | 73.9 (71.0-78.7)        | 73.6 (70.2-78.7)        |
| Bold values indicate the best performing results for each evaluation metric across compared models. |             |        |     |                         |                         |                         |                         |



**FIGURE 8.** Performance comparison of different fusion strategies in corneal topography classification.

features. Particularly in the Sag-A mode, CMVFT, leveraging the long-distance modeling advantage of transformers, successfully captures the specific contours of the lesion's tongue-like structure, further validating its sensitivity to complex pathological features.

The CMVFT model demonstrated precise attention to the pathologically relevant areas in all input modes, with its activation maps closely aligning with clinically important regions. This indicates that its understanding of the lesion core is significantly superior to that of other models. This characteristic underscores the clinical application potential of CMVFT, particularly in early screening, progres-

sion monitoring, and treatment planning for KCN, which is of significant value.

- **ABLATION STUDY:** To evaluate the contribution of each component in the corneal topography three-classification task, a series of ablation experiments were designed. The results of these experiments are shown in Table 5. The experiments began with a baseline model and progressively introduced the attention mechanism, fusion module (Fusion-T), and MSAM to analyze the impact of each component on classification performance. The results indicate that using the attention mechanism in feature fusion optimizes

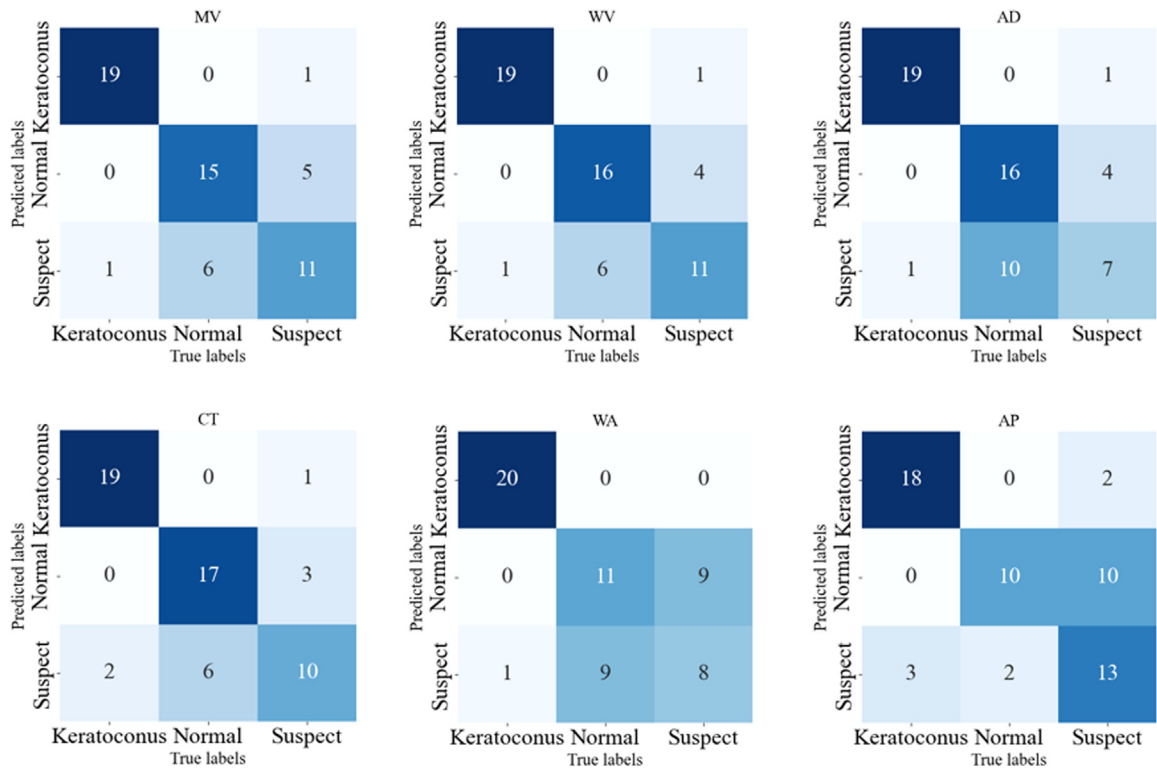


FIGURE 9. Comparison of confusion matrices for different fusion methods in multiview abnormal corneal detection task.

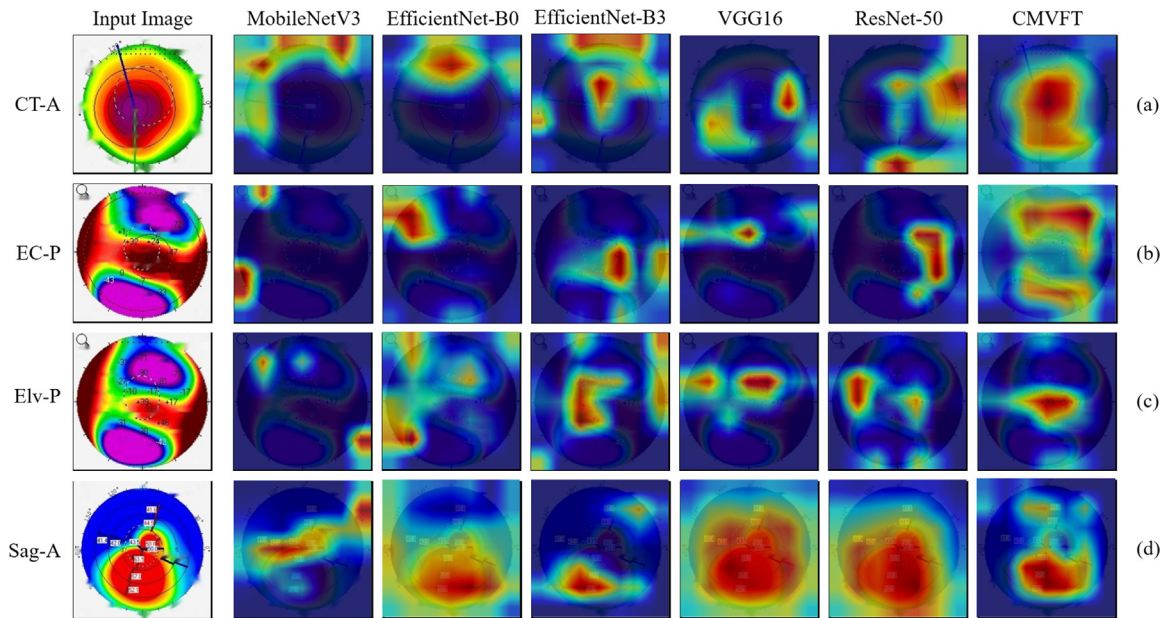


FIGURE 10. Comparison of Grad-CAM maps for four refractive topography models in KCN samples.

the model's classification performance. Further, the introduction of the Fusion-T module significantly improved the model's performance. Finally, incorporating the multiscale feature module led to the best performance across all evaluation metrics. These results validate the important role of each component in enhancing the model's classification

ability, especially the critical role played by the fusion module in integrating multiview information.

• **EXPERIMENTAL RESULTS ANALYSIS AND DISCUSSION:** The Transformer-based multiview fusion framework CMVFT proposed in this study demonstrated significant

**TABLE 5.** Ablation Study Results for Corneal Topography Classification Performance Based on Different Components

|          | Components |      |          | Disease Classification |              |              |              |
|----------|------------|------|----------|------------------------|--------------|--------------|--------------|
|          | Attention  | MSAM | Fusion-T | Precision              | Acc          | Recall       | F1           |
| Baseline |            |      |          | 0.658                  | 0.673        | 0.672        | 0.668        |
|          | ✓          |      |          | 0.707                  | 0.713        | 0.707        | 0.708        |
|          |            |      | ✓        | 0.752                  | 0.759        | 0.759        | 0.756        |
|          |            | ✓    | ✓        | <b>0.821</b>           | <b>0.828</b> | <b>0.828</b> | <b>0.824</b> |

Bold values indicate the best performing results for each evaluation metric across compared models.

performance improvement in the corneal topography three-class classification task, particularly excelling in the classification performance of the “Suspect” category, which significantly outperformed traditional methods. The experimental results showed that CMVFT achieved optimal levels in multiple metrics, including AUC, Precision, and F1, with the AUC for the “Suspect” category reaching 86.3%, much higher than that of other models. This indicated that the multiview fusion approach, combined with the global modeling capability of Transformer, could effectively differentiate complex categories even in the presence of feature overlap. Compared to convolutional neural networks, Transformer captured global interaction features across multiple views, and through the multiscale attention module, it extracted fine-grained local and global feature information, further enhancing classification robustness.

In contrast to traditional convolutional networks, Transformer also demonstrated greater scalability, with performance significantly improving as the amount of data increases, while convolutional networks tend to experience performance bottlenecks with larger datasets. This scalability is especially important in medical image analysis tasks, where this feature of the Transformer architecture can enhance diagnostic performance through dataset expansion in clinical applications.<sup>46,47</sup> However, the advantages of Transformer also came with certain challenges, such as higher computational complexity and dependence on hardware resources, which limited its application in scenarios requiring real-time processing. To address this, we reduced the computational load by performing hierarchical fusion on some views.

To evaluate the real-world clinical utility of our system, we computed expected PPVs for KCN and Suspect classifications using representative prevalence data from population studies. In the Rotterdam Study (Netherlands), the prevalence of KCN was 0.38%, and that of Suspect KCN was 2.71% (N = 2660).<sup>48</sup> Using our model’s KCN sensitivity (100%) and specificity (94.7%), the projected PPV is 6.7%, indicating that approximately 1 in 15 positive predictions corresponds to a true case in a general adult population. For Suspect KCN, with sensitivity (72.2%) and specificity (90.0%), the expected PPV is 16.9%. In the Yamagata Study (Japan), where KCN prevalence was 0.85% and

Suspect prevalence 1.46%, the expected PPVs increase to 13.8% for KCN and 11.0% for Suspect cases, respectively.<sup>49</sup> These findings suggest that although PPVs remain modest due to low prevalence, our model can serve as an effective triage tool, substantially enriching the pool of patients referred for further diagnostic evaluation.

Although the multiview fusion strategy exhibited a significant overall performance advantage, the model’s dependency on seven views could lead to issues such as information redundancy and noise interference. The experiment did not systematically assess the individual contribution of each view, making it unclear how each view contributed to the improvement in classification performance. Specifically, for the average classification performance, CMVFT’s AUC was slightly lower than ResNet-152 (90.9% vs 91.8%), which could be related to redundant features or noise between views. Although the introduction of Transformer improved performance, its learning capability for complex samples still had certain limitations. In particular, the classification performance was still affected when the feature space between the “Normal” and “Suspect” categories overlapped significantly. Without increasing the dataset size, this study has demonstrated the feasibility of small-sample learning in corneal topography tasks, and this direction remained an area for further exploration. Additionally, due to the relatively limited and single-source dataset, the results of this study may have certain biases, and the generalization performance under different devices and data distributions requires further validation.

## CONCLUSION

This study proposes the Corneal Multiview Fusion Transformer (CMVFT) as a novel framework to tackle the three-class classification task of corneal topography. By custom-designing a multiscale attention module to bridge the representational gap between convolutional inductive biases and Transformer-based long-range modeling, CMVFT provides a smooth transition from local feature extraction to global context analysis. This design fundamentally addresses the bottleneck of detecting the “Suspect” category—where feature overlap between normal and KCN often confounds tra-



ditional CNN-based methods. In doing so, CMVFT leverages seven standard corneal views without relying on naive concatenation or single-view analysis, thus fully exploiting cross-view interactions while maintaining superior generalization in small-sample scenarios. Experimental results confirm significant gains in key metrics (AUC, Precision, Recall, F1), particularly for the “Suspect” category, underscoring CMVFT’s unprecedented classification performance.

Furthermore, class-specific PPV analysis under real-world prevalence conditions from the Yamagata Study (Japan) further validates the model’s clinical utility. The proposed CMVFT model achieves higher PPVs for both KCN (13.8% vs 13.1%) and Suspect cases (11.0% vs 10.5%) compared to previous strategies, while maintaining strong recall and balanced specificity. These results demonstrate that CMVFT not only improves label-level diagnostic precision but also enhances the reliability of AI-assisted early keratoconus screening in population settings.

**Future Work.** Moving forward, expanding the dataset size and diversity (eg, different devices, ethnicities, disease stages) will enhance CMVFT’s robustness. A systematic contribution analysis of each corneal view can further optimize view selection strategies to reduce redundancy and computational burden. Additionally, lightweight

Transformer variants are worth exploring to accommodate real-time diagnostic environments with limited resources. Integrating this framework into clinical decision support systems may bolster preoperative screening and continuous disease monitoring. Finally, embedding interpretability techniques such as attention-weight visualization or Grad-CAM could increase clinician trust and adoption. Overall, CMVFT offers important insights into multiview fusion and Transformers for medical image analysis, with substantial promise for broader applications in complex classification tasks that support precision medicine and personalized healthcare.

## CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

**Yifan Lu:** Writing – review & editing, Writing – original draft, Methodology. **Baojiang Li:** Writing – review & editing, Supervision, Conceptualization. **Yunhai Zhang:** Writing – review & editing, Validation, Investigation. **Yayu Qi:** Writing – review & editing, Resources, Investigation. **Xingbin Shi:** Writing – review & editing, Resources, Investigation.

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